

- I leave the handin open, for late turn in

- Lab handin

- only 1 person from each group does handin

- copy a file to the unix1.csc.calpoly.edu → use Filezilla

- to see what directories are open: handin jseng

- to handin a file: handin jseng CPE315_lab1 mod.asm

- to check what files are handed in: handin jseng CPE315_lab1

/home/jseng/.snapshot

-no lab
today

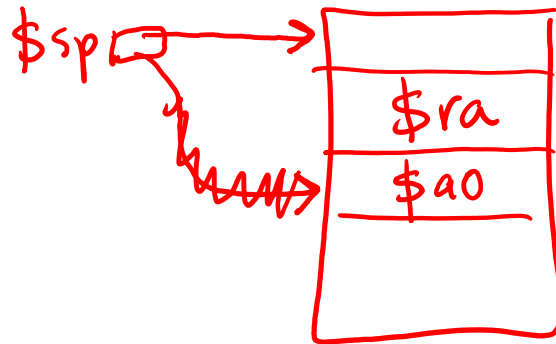
$\$a0$ ↓
Sum (int n) {
 if (n == 0)
 return 0;
 else
 return Sum(n-1) + n;
}

base case →

$\$a0$ ↓
saves n to the stack

decrement n

$\$v0$



Sum:
beg $\$a0, \0 , base_case
addi $\$sp, \$sp, -8$
sw $\$a0, 0(\$sp)$
sw $\$ra, 4(\$sp)$
addi $\$a0, \$a0, -1$
jal Sum
lw $\$a0, 0(\$sp)$
lw $\$ra, 4(\$sp)$
addi $\$sp, \$sp, 8$
add $\$v0, \$v0, \$a0$
base_case: jr $\$ra$
add $\$v0, \$0, \$0$
jr $\$ra$

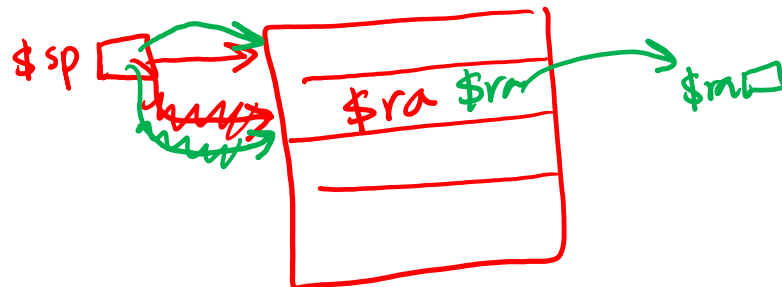
- Lab 1 progress
check today
- no class 4/23 - 4/27

```

int count_wheels(int c) {
    base-case { if (c == 0)
                return 0;
            }
    else
        return count_wheels(c-1) + 4;
}

```

Annotations: $\$a0$ points to c . $\$v0$ points to the return value. $\$v0$ is also under the $c-1$ in the recursive call.



$\$a0 = 100$

count_wheels:

beg $\$a0, \0 , base-case

push $\$ra$ to stack {

```

    addi $sp, $sp, -4
    sw $ra, 0($sp)
    addi $a0, $a0, -1
    jal count_wheels
    addi $v0, $v0, 4
    lw $ra, 0($sp)
    addi $sp, $sp, 4
}

```

pop $\$ra$ off stack {

base-case: jr $\$ra$

```

    addi $v0, $0, 0
    jr $ra

```